

Homework

You must complete problems 1 through 3 and then choose 1 of either problem 4 or 5. If you complete both 4 and 5, you will get a max of 5 points added onto your exams (based on problem correctness).

All text responses must be written by hand (problems 1 through 3); you may use electronic mediums such as a tablet or write them on paper and take a picture of it. All writing must be neat, organized, and legible. Workflow section problems will require a plot from software such as excel or python, and your code must be uploaded separately. Submissions will be uploaded to Blackboard. You may work with others, but your final work must be your own.

Section 1: Theoretical Concepts

Question 1 (CA): Draw a labelled diagram of a SCF loop in DFT. Then draw a second diagram of how this SCF loop fits into a geometry optimization loop. Explain both loops in 3-4 sentences outlining what each portion of the algorithm does.

Question 2 (CA): How are forces calculated in DFT (be specific and mathematical in your description) and does the force calculation method depend on the wavefunction representation? If so, why (be specific)?

Question 3 (CA, CC): In class we discussed LDA and GGA exchange-correlation functionals. Provide labelled mathematical representation for meta-GGA, hybrid, and RPA exchange-correlation functionals and discuss their accuracy-vs-computational cost. Also, discuss which exchange-correlation functionals can you reliably use for each material classes (metals, ceramics, and polymers/molecules) and explain why.

Section 2: Problem Solving

Question 4 (CA, CB): A geometry optimization uses electronic-structure calculations to determine the forces acting on atoms and then iteratively moves the atoms toward a lower-energy configuration. In VASP, each ionic step contains one or more self-consistent electronic calculations followed by a structural update based on the calculated forces.

For this problem, you will begin from a deliberately distorted diamond-silicon structure and use VASP to relax the atomic positions while keeping the simulation cell fixed. You will then examine how the total energy, atomic forces, and atomic coordinates evolve during the optimization.

Supplied files:

data/problem4/POSCAR - distorted silicon structure

data/problem4/POTCAR - silicon PAW potential

*You will be expected to make your own INCAR (see below) and KPOINTS file for VASP.

Your task

A. Prepare the VASP calculation

1. Create a working directory for the geometry optimization and copy all supplied files over to it.
2. Create an INCAR file for a fixed-cell ionic relaxation. Use the following settings as a starting point:

SYSTEM = Distorted Si geometry optimization

ENCUT = 400

EDIFF = 1E-6

EDIFFG = -0.02

IBRION = 2

ISIF = 2

NSW = 30

PREC = Accurate

LREAL = Auto

3. Submit the calculation to the SCC using the VASP batch script provided for the course (recall first day of class and course materials layout on the SCC).
4. Confirm that the calculation reaches the requested force criterion or otherwise determine why it stopped.

B. Compare the starting and final structures

1. Compare the original POSCAR with the final CONTCAR.

2. Identify which atom or atoms moved most during the relaxation.
3. Calculate a radial distribution function for each structure using python.
4. Report the largest atomic displacement in Å.

C. Examine the ionic optimization history

1. Use a Python script to extract the total energy for every ionic step from the VASP output.
2. Extract the maximum force magnitude at every ionic step.
3. Report the total number of ionic steps performed.
4. Report the initial and final total energies and the final maximum force.

D. Visualize the relaxation

Create one figure containing two plots or two separate saved figures:

Total energy versus ionic step.

Maximum atomic force versus ionic step.

Include axis labels with units and descriptive titles. Save the final figure as `problem4_geometry_optimization.png`.

E. Connect the electronic and ionic loops

Inspect the OSZICAR or OUTCAR for at least two different ionic steps. For each ionic step, identify that VASP performs multiple electronic SCF iterations before moving the atoms. Briefly report approximately how many electronic iterations were required near the beginning of the relaxation and near the end.

F. Interpret the workflow

In approximately 150-250 words, answer the following:

1. Why must the electronic structure be converged before VASP can reliably update the atomic positions?
2. Why does the total energy generally decrease as the geometry optimization proceeds?
3. Why is a force-based stopping criterion such as EDIFFG more physically useful for geometry optimization than simply stopping when the total energy changes only slightly?
4. Based on the final forces and RDF, would you consider the final structure relaxed to its crystalline ground state structure? Explain.

What to submit

1. Your INCAR file and Python analysis script.
2. problem4_geometry_optimization.png.
3. A small table containing the number of ionic steps, initial energy, final energy, final maximum force, and largest atomic displacement.
4. Your 150-250 word interpretation.
5. Plots of RDF for each structure, total energy versus ionic step, and maximum atomic force versus ionic step.

Important: You are not graded on obtaining a perfectly converged research-quality structure. You are graded on correctly constructing the geometry-optimization workflow, extracting the relevant convergence information, and interpreting what the calculation is doing.

Question 5 (CA, CB, CC): The electronic density of states (DOS) describes how many electronic states are available within a given energy interval. The behavior of the DOS near the Fermi level provides a direct way to distinguish a metal from a semiconductor or insulator. A metal has accessible electronic states at the Fermi level, while a semiconductor has an energy range with no allowed states between the valence and conduction bands.

For this problem, you will calculate and compare the electronic density of states of two simple crystalline materials: FCC aluminum and diamond silicon. The systems are intentionally small so that the calculations can be completed quickly on the SCC.

Supplied files:

data/problem5/aluminum/POTCAR - aluminum PAW potential

data/problem5/silicon/POTCAR - silicon PAW potential

*You will need to make a POSCAR for Al and you should use your relaxed Si CONTCAR (copy any rename it to POSCAR-VASP requires the input to be called POSCAR).

Your task

A. Perform self-consistent calculations

For both materials, first perform a standard self-consistent calculation using a regular Monkhorst-Pack or Gamma-centered k-point mesh. You may use the KPOINTS file from problem 4 here.

1. Create separate calculation directories for Al and Si.
2. Use the same general convergence settings for both materials. A suitable starting INCAR is:
ENCUT = 400
EDIFF = 1E-6
PREC = Accurate
LREAL = Auto
LWAVE = .TRUE.
LCHARG = .TRUE.
3. Use a reasonably dense k-point mesh appropriate for these small primitive cells. State the mesh you used for each material.
4. Submit both calculations to the SCC and confirm that both electronic SCF calculations converge.

B. Perform DOS calculations

After the SCF calculation has converged, perform a second calculation for each material using the converged electronic density and a denser k-point mesh to obtain a smoother DOS.

1. Use the converged charge density from the SCF calculation, same structures.

2. Use a DOS energy grid fine enough to resolve the states near the Fermi level. A suitable starting point is NEDOS = 2000.

3. Use a small smearing width appropriate for visualizing the DOS. State the ISMEAR and SIGMA values you used.

C. Extract and align the DOS

1. Write a Python program that reads the VASP DOS output for both materials.

2. Shift the energy axis so that the Fermi energy is at 0 eV for each material:

$$E_{\text{plot}} = E - E_F$$

You can get the Fermi energy by using the following command in a linux terminal where the OUTCAR is: `grep "E-fermi" OUTCAR`. You should see something in the terminal such as:

E-fermi : 5.4321 XC(G=0): ...

The 5.4321 would be the Fermi energy.

3. Extract the total DOS as a function of energy for both Al and Si.

4. Report the Fermi energy obtained from each calculation before shifting the energy axis.

D. Compare metallic and semiconducting behavior

1. Create a figure with the Al and Si DOS shown in separate panels or clearly separated curves.

2. Mark the Fermi level at 0 eV.

3. For aluminum, determine whether the DOS at the Fermi level is approximately zero or nonzero.

4. For silicon, identify the valence-band edge and conduction-band edge from the DOS and estimate the calculated band gap.

5. Compare the calculated silicon band gap with the experimental room-temperature value of approximately 1.1 eV.

Save the final figure as problem5_Al_vs_Si_DOS.png.

E. Create a concise comparison table

Create a table containing at least the following quantities:

Material	Fermi energy (eV)	DOS at EF	Calculated gap (eV)	Electronic classification
Al				

Si				
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F. Interpret the result

In approximately 150-250 words, answer the following:

1. What feature of the DOS near the Fermi level distinguishes aluminum from silicon?
2. Why can aluminum conduct electrically without electrons first being excited across an energy gap?
3. How does the calculated silicon band gap compare with experiment?
4. Would the DOS alone be sufficient to determine whether an unknown material is metallic or semiconducting? Explain what feature you would examine.

What to submit

1. Your VASP INCAR and KPOINTS files for both materials and your Python DOS-analysis script.
2. problem5_Al_vs_Si_DOS.png.
3. Your Al-versus-Si comparison table.
4. Your 150-250 word interpretation.

Important: You are not graded on reproducing the experimental silicon band gap exactly. You are graded on correctly performing the SCF-to-DOS workflow, aligning and interpreting the DOS, distinguishing metallic from semiconducting behavior, and explaining the limitations of the calculation.